Week 2 Homework

This is a preview of the published version of the quiz

Started: Jul 2 at 7:47am

Quiz Instructions

Please answer all the questions below.

Question 1

1 pts

(Lesson 2.5: Probability Basics.) If $P(A) = P(B) = P(C) = 0.6$ and $A, B,$ and C are independent, find the probability that exactly one of $A, B,$ and C occurs.

- a. 0.144
- b. 0.288
- c. 0.576
- d. 0.6
- e. I'm from The University Of Georgia. Is the answer -3?

Question 2

1 pts

(Lesson 2.5: Probability Basics.) Toss 3 dice. What's the probability that a "4" will come up exactly twice?

- a. 5/72
- b. 1/2
- c. 13/16
- d. 1/8

Question 3

1 pts

(Lesson 2.7: Great Expectations.) Suppose that X is a discrete random variable having $X = -1$ with probability 0.2, and $X = 3$ with probability 0.8. Find $\mathbf{E}[X]$.

- a. -1
- b. 3
- c. 1
- d. 2.2

Question 4

1 pts

(Lesson 2.7: Great Expectations.) Suppose that X is a discrete random variable having $X = -1$ with probability 0.2, and $X = 3$ with probability 0.8. Find $\mathbf{Var}[X]$.

- a. -1
- b. 1
- c. 2.56
- d. 5.12

Question 5

1 pts

(Lesson 2.7: Great Expectations.) Suppose that X is a discrete random variable having $X = -1$ with probability 0.2, and $X = 3$ with probability 0.8. Find $\mathbf{E}[3 - \frac{1}{X}]$.

- a. 3

- b. ∞
- c. -2
- d. 44/15

Question 6

1 pts

(Lesson 2.7: Great Expectations.) Suppose X is a continuous random variable with p.d.f. $f(x) = 4x^3$ for $0 \leq x \leq 1$. Find $\mathrm{E}[1/X^2]$.

- a. 2/3
- b. 1
- c. 3/2
- d. 2

Question 7

1 pts

(Lesson 2.8: Functions of a Random Variable.) Suppose X is the result of a 5-sided die toss having sides numbered $-2, -1, 0, 1, 2$. Find the probability mass function of $Y = X^2$.

- a. $\mathrm{P}(Y = 1) = \mathrm{P}(Y = 4) = 1/2$
- b. $\mathrm{P}(Y = 1) = \mathrm{P}(Y = 2) = 1/2$
- c. $P(Y = 0) = \frac{1}{5}, \; and \; P(Y = 1) = P(Y = 4) \; = \frac{2}{5}$
- d. $\mathrm{P}(Y = -2) = \mathrm{P}(Y = -1) = \mathrm{P}(Y = 0) = \mathrm{P}(Y = 1) = \mathrm{P}(Y = 2) = 1/5$

Question 8

1 pts

(Lesson 2.8: Functions of a Random Variable.) Suppose X is a continuous random variable with p.d.f. $f(x) = 2x$ for $0 < x < 1$. Find the p.d.f. $g(y)$ of $Y = X^2$. (This may be easier than you think.)

- a. $g(y) = 1$, for $0 < y < 1$
- b. $g(y) = y$, for $0 < x < 1$
- c. $g(y) = y^2$, for $-1 < y < 1$
- d. $g(y) = x^2$, for $0 < y < 1$

Question 9

1 pts

(Lesson 2.9: Jointly Distributed RVs.) Suppose that $f(x, y) = 6x$ for $0 \leq x \leq y \leq 1$. Find $\mathrm{P}(X < 1/2 \text{ and } Y < 1/2)$.

- a. 1
- b. 1/2
- c. 1/4
- d. 1/8

Question 10

1 pts

(Lesson 2.9: Jointly Distributed RVs.) Suppose that $f(x, y) = 6x$ for $0 \leq x \leq y \leq 1$. Find the marginal p.d.f. $f_X(x)$ of X.

- a. $6x(1 - x)$, for $0 \leq x \leq 1$
- b. $6x$, for $0 \leq x \leq 1$
- c. $6y$, for $0 \leq x \leq 1$
- d. $6x(1 - y)$, for $0 \leq x \leq 1$

Question 11

1 pts

(Lesson 2.9: Jointly Distributed RVs.) YES or NO? Suppose X and Y have joint p.d.f. $f(x, y) = cxy/(1 + x^2 + y^2)$ for $0 < x < 1$, $0 < y < 1$, and whatever constant c makes the nasty mess integrate to 1. Are X and Y independent?

- a. Yes
- b. No

Not saved

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